

Dynamics and Reconstruction from Observation

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Abstract

Starting from a σ -additive probability measure on an observable algebra [5], this paper derives two structures that are often assumed: the dynamics of prediction, and the recoverability of the state space.

Making prediction precise forces the introduction of the predictive kernel $\Pi_Q : \alpha \rightarrow \mathcal{P}(\beta)$, the regular conditional distribution of a future observation F given a current observation Q . Composing with Q gives the minimal predictive state map $Q_* : \Omega \rightarrow \mathcal{P}(\beta)$. Indexing over time, the kernels compose via Chapman–Kolmogorov — derived, not assumed — yielding a Markov semigroup $\{K_t\}$ and the Koopman–Perron duality $\int K_t g d\mu = \int g d(\mu \star \Pi_t)$. When the kernels are Dirac measures, K_t collapses to the classical Koopman operator.

The construction produces a predictive state space $\mathcal{P}(\beta)$. In a measure-preserving system $(X, \mathcal{B}, \mu, T)$ with $Q_t = h \circ T^t$, the map Q_* becomes the delay map $\Phi_h(x) = (h(T^n x))_{n \in \mathbb{Z}}$. The reconstruction question is then internal to the same framework: whether this predictive state is faithful — whether the prediction machinery sees all of X , or only a proper quotient.

The answer is a three-way equivalence — the reconstruction theorem. State-space recovery ($\mathcal{O}_h = \mathcal{B} \bmod \mu$) holds if and only if the delay algebra $\mathcal{A}_h$ is L^2 -dense, if and only if Φ_h is a measure-theoretic embedding. The bridge between the first two conditions is a general lemma: for any subalgebra $\mathcal{A} \subseteq L^\infty$ containing 1, the L^2 -closure of $\mathcal{A}$ equals $L^2(X, \sigma(\mathcal{A}), \mu)$. When reconstruction holds, the compact Stone space [5] — built as a completion of the sample space for deriving σ -additivity — is identified with X itself.

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1 Introduction

The classical approach to dynamics, following Koopman [3], describes how observable functions evolve under a known transformation T : the operator $U_T f = f \circ T$ acts by precomposing with T as a primitive. This paper asks two questions that precede this picture.

The first is: given a current observation $Q : \Omega \rightarrow \alpha$ and a future observation $F : \Omega \rightarrow \beta$, what does an observer predict about F given Q ? The dynamics, if there are any, should follow from the answer — not precede it. Making prediction precise immediately requires a σ -additive probability measure on the observable algebra. That measure is not an extra assumption: a coherent family of finitely additive charges on a directed system of observable algebras extends to a genuine probability measure exactly when it satisfies collective exhaustion [5], the condition ruling out persistent mass on globally vanishing events.

Once the measure exists, the canonical answer to the prediction question is the regular conditional distribution of F given Q : the unique Markov kernel $\Pi_Q : \alpha \rightarrow \mathcal{P}(\beta)$ realising $P(F \in \cdot \mid Q = q)$ simultaneously across all outcomes. Its existence rests on the Rokhlin disintegration theorem [6], which requires σ -additivity. Composing with Q gives the minimal predictive state map $Q_* : \Omega \rightarrow \mathcal{P}(\beta)$, the minimal sufficient statistic for future prediction. Indexing over time,

the kernels $\{\Pi_t\}$ compose; their composition rule is the Chapman–Kolmogorov equation, derived from temporal coherence rather than assumed. From it emerge a semigroup of Markov operators $\{K_t\}$ and the Koopman–Perron duality connecting them with the pushforward operators. When the kernels are Dirac measures, the construction collapses to the classical Koopman picture.

The construction produces a predictive state space $\mathcal{P}(\beta)$ and a stationary distribution ν_{Q_*} . In a measure-preserving system $(X, \mathcal{B}, \mu, T)$ with $Q_t = h \circ T^t$ for some $h \in L^\infty$, the map Q_* specialises to the delay map $\Phi_h(x) = (h(T^n x))_{n \in \mathbb{Z}}$. The reconstruction question is therefore not a separate problem: it asks whether this predictive state is faithful — whether the prediction machinery sees all of X , or only a proper quotient. The answer is a three-way equivalence: the σ -algebraic condition $\mathcal{O}_h = \mathcal{B}$ modulo μ , the analytic condition that the delay algebra $\mathcal{A}_h$ is L^2 -dense, and the geometric condition that Φ_h is a measure-theoretic embedding are all equivalent.

The key to the equivalence is the density bridge lemma: for any subalgebra $\mathcal{A} \subseteq L^\infty(X, \mu)$ containing the constant 1, the L^2 -closure of $\mathcal{A}$ equals $L^2(X, \sigma(\mathcal{A}), \mu)$. The proof goes via the functional monotone class theorem [1]. This is what lets the σ -algebraic condition ($\mathcal{O}_h = \mathcal{B} \bmod \mu$) and the analytic condition ($\mathcal{A}_h$ L^2 -dense) pass freely between each other — they are not two separate conditions but two descriptions of the same constraint.

When reconstruction holds, the Stone space [5] $\text{St}(\mathcal{O}_h)$ — built as a compact completion of the sample space for deriving σ -additivity — turns out to be measure-theoretically isomorphic to X itself. The programme that began by seeking a compact ambient space for the measure ends by identifying it with the state space being observed.

The paper is self-contained modulo the probability measure. Sections 2 and 3 develop the predictive structure from a probability space $(\Omega, \mathcal{F}, P)$. Section 4 makes the connection explicit: specialising Q_* to a measure-preserving system yields the delay map, and the reconstruction question becomes whether that map is injective modulo null sets. Sections 5 and 6 carry out the reconstruction, proving the density bridge and the reconstruction theorem.

2 Prediction and the minimal predictive state

2.1 Predictive kernel

Fix a probability space $(\Omega, \mathcal{F}, P)$ and two measurable maps:

- $Q : \Omega \rightarrow \alpha$, where $(\alpha, \mathcal{A})$ is a measurable space;
- $F : \Omega \rightarrow \beta$, where $(\beta, \mathcal{B})$ is a standard Borel space.

The standard Borel hypothesis on β is used once: it is the hypothesis under which the Rokhlin disintegration theorem [6] guarantees the existence of a regular conditional distribution. All other results hold in greater generality.

We write $\mathcal{P}(\beta)$ for the space of probability measures on β , equipped with its canonical (Giry) σ -algebra. The joint pushforward $\rho := P \circ (Q, F)^{-1}$ on $\alpha \times \beta$ has marginals $\rho_\alpha = P \circ Q^{-1}$ and $\rho_\beta = P \circ F^{-1}$.

The regular conditional distribution of F given Q is the unique (up to $P \circ Q^{-1}$ -null sets) Markov kernel $\Pi_Q : \alpha \rightarrow \mathcal{P}(\beta)$ satisfying

$$\int_A \Pi_Q(q, B) d(P \circ Q^{-1})(q) = P(Q \in A, F \in B)$$

for all measurable $A \subseteq \alpha$ and $B \subseteq \beta$.

Definition 2.1 (Predictive kernel). The *predictive kernel* is Π_Q , the disintegration kernel of $\rho = P \circ (Q, F)^{-1}$ over its first marginal: $\Pi_Q(q, A) = P(F \in A \mid Q = q)$.

The kernel is covariant under measurable maps: if $\pi : \alpha \rightarrow \alpha'$ is measurable and $Q' = \pi \circ Q$, then $\Pi_Q(q, \cdot) = \Pi_{Q'}(\pi(q), \cdot)$ for $(P \circ Q^{-1})$ -almost every q , by uniqueness of disintegration.

Composing Π_Q with Q gives the map sending each sample point to its conditional law.

2.2 Minimal predictive state

Definition 2.2 (Minimal predictive state map). The *minimal predictive state map* is

$$Q_* : \Omega \rightarrow \mathcal{P}(\beta), \quad Q_*(\omega) = \Pi_Q(Q(\omega), \cdot).$$

Since Π_Q is a Markov kernel it is measurable, and Q_* inherits measurability by composition.

Theorem 2.3 (Predictive factorization). *For every bounded measurable $g : \beta \rightarrow \mathbb{R}$:*

$$E[g(F) \mid \sigma(Q)] = \int g dQ_*(\cdot) \quad P\text{-a.e.}$$

Proof. Let $h(\omega) := \int g dQ_*(\omega)$. We verify the two defining properties.

Measurability. The map $q \mapsto \int g d\Pi_Q(q, \cdot)$ is strongly measurable; since $Q_* = \Pi_Q(Q(\cdot), \cdot)$, the map h is $\sigma(Q)$ -measurable.

Set-integral identity. For $S = Q^{-1}(A)$:

$$\begin{aligned} \int_S h dP &= \int_A \int g d\Pi_Q(q, \cdot) d(P \circ Q^{-1})(q) \quad (\text{pushforward}) \\ &= \int_{A \times \beta} g(f) d\rho(q, f) \quad (\text{disintegration}) \\ &= \int_S g(F) dP. \end{aligned}$$

The conclusion follows from a.e. uniqueness of conditional expectation. $\square$

Corollary 2.4 (Predictive sufficiency). $E[g(F) \mid \sigma(Q)] = E[g(F) \mid \sigma(Q_*)]$ P -a.e.

Proof. $\sigma(Q_*) \subseteq \sigma(Q)$, and the conditional expectation given $\sigma(Q)$ is already $\sigma(Q_*)$ -measurable by the theorem. $\square$

The operator $(K_{Q_*}g)(q_*) = \int g dq_*$ on $\mathcal{P}(\beta)$ recasts the theorem as: $E[g(F) \mid \sigma(Q)] = (K_{Q_*}g) \circ Q_*$ P -a.e.

3 Dynamics and duality

3.1 Markov semigroup

To introduce dynamics, index over a discrete time horizon. For each $t \in \mathbb{N}$, let $\Pi_t : \gamma \rightarrow \mathcal{P}(\gamma)$ be a Markov kernel on the state space γ , with $\Pi_0(q, \cdot) = \delta_q$. A family $\{\Pi_t\}$ satisfying the Chapman–Kolmogorov equation

$$\Pi_{t+s}(q, \cdot) = \int_{\gamma} \Pi_t(q', \cdot) d\Pi_s(q, q') \quad (\text{i.e., } \Pi_{t+s} = \Pi_t \circ_{\kappa} \Pi_s)$$

is a *Markov semigroup kernel*. The associated *Markov operators* are

$$(K_t g)(q) := \int g d\Pi_t(q, \cdot), \quad g : \gamma \rightarrow \mathbb{R}.$$

Each K_t is positive, unital, and contractive on bounded measurable functions, with $K_0 = \text{id}$; positivity, unitality, and contractiveness are immediate from the Markov property.

Proposition 3.1 (Semigroup property). $K_{t+s}g = K_t(K_sg)$ for all bounded measurable g .

Proof. $(K_{t+s}g)(q) = \int g d(\Pi_t \circ_\kappa \Pi_s)(q, \cdot) = \int \left(\int g d\Pi_t(q', \cdot) \right) d\Pi_s(q, q') = \int (K_tg)(q') d\Pi_s(q, q') = (K_t(K_sg))(q)$, where the exchange of integrals is justified by Bochner–Fubini. $\square$

The pushforward of a measure μ on γ through the kernel is $P_t^*\mu := \mu \star \Pi_t = \int \Pi_t(q, \cdot) d\mu(q)$.

3.2 Koopman–Perron duality

Theorem 3.2 (Koopman–Perron duality). *For every bounded measurable g and finite measure μ :*

$$\int (K_tg) d\mu = \int g d(P_t^*\mu).$$

Proof. $\int g d(P_t^*\mu) = \int g d(\mu \star \Pi_t) = \int_\gamma \int g d\Pi_t(q, \cdot) d\mu(q) = \int (K_tg) d\mu$, by Bochner–Fubini. $\square$

When $\Pi_t(q, \cdot) = \delta_{\varphi_t(q)}$ for a measurable map $\varphi_t : \gamma \rightarrow \gamma$, the stochastic picture collapses.

3.3 Specialisation to measure-preserving systems

Proposition 3.3 (Deterministic limit). *If $\Pi_t(q, \cdot) = \delta_{\varphi_t(q)}$, then $K_tg = g \circ \varphi_t$ and $\varphi_{t+s} = \varphi_t \circ \varphi_s$.*

Proof. $(K_tg)(q) = \int g d\delta_{\varphi_t(q)} = g(\varphi_t(q))$. The flow law follows from $\delta_{\varphi_{t+s}(q)} = (\Pi_t \circ_\kappa \Pi_s)(q, \cdot) = \delta_{\varphi_t(\varphi_s(q))}$ and injectivity of the Dirac embedding. $\square$

In the deterministic case, $K_tg = g \circ \varphi_t = U_T^t g$: the Markov operator specialises to the classical Koopman operator. The Koopman–Perron duality $\int K_tg d\mu = \int g d(P_t^*\mu)$ becomes the change-of-variables formula $\int g \circ T^t d\mu = \int g d(T_*^t\mu)$.

Example 3.4 (Finite Markov chain). Let $X = \{1, \dots, k\}$, $\mathcal{B} = 2^X$, and let $P = (p_{ij})$ be the transition matrix of a finite irreducible Markov chain with unique stationary distribution μ ($\mu P = \mu$). Take full-state observation $Q_t = X_t$. The predictive kernel is $\Pi(i, \cdot) = P(i, \cdot)$, the i -th row of P : this is the unique Markov kernel satisfying $\mathbb{E}_\mu[f(X_1) \mid X_0 = i] = \sum_j p_{ij}f(j)$ for all bounded f . The t -step kernel is $\Pi^{(t)}(i, \cdot) = P^t(i, \cdot)$, and temporal coherence gives $\Pi^{(t+s)} = \Pi^{(t)} \circ_\kappa \Pi^{(s)}$, i.e. $P^{t+s} = P^t P^s$. The semigroup acts as

$$K_t f(i) = \sum_j p_{ij}^t f(j).$$

Koopman–Perron duality reduces to the stationary-measure identity:

$$\int K_tg d\mu = \mu^\top P^t g = (\mu P^t)^\top g = \mu^\top g = \int g d\mu,$$

the fixed-point property $\mu P = \mu$ of the stationary distribution. In this finite setting every charge is automatically σ -additive, so the admissibility condition of [5] is trivially satisfied. Reconstruction holds at lag 0: $\mathcal{O}_h = \sigma(h) = \mathcal{B}(X)$ whenever h separates points of X .

The full structure produced by the prediction construction is captured in one definition.

Definition 3.5 (Observable dynamical system). An *observable dynamical system* is a triple $(\mathcal{P}(\beta), \nu_{Q_*}, \{K_t\}_{t \in \mathbb{N}})$ where $\mathcal{P}(\beta)$ is the predictive state space (the image of Q_*), $\nu_{Q_*} = P \circ Q_*^{-1}$ is the pushforward of P along Q_* , and $\{K_t\}$ is the Markov semigroup on $\mathcal{P}(\beta)$.

4 From prediction to reconstruction

Setting $Q_t = h \circ T^t$ in a measure-preserving system $(X, \mathcal{B}, \mu, T)$ with $h \in L^\infty$ specialises the construction. Under this choice, $Q_*(\omega) = (h(T^n \omega))_{n \in \mathbb{Z}}$ is the delay map $\Phi_h(x) = (h(T^n x))_{n \in \mathbb{Z}}$, and the observable dynamical system is the triple $(\Phi_h(X), \mu \circ \Phi_h^{-1}, \{K_t\})$.

The predictive state space $\Phi_h(X)$ is a subset of $\mathbb{R}^{\mathbb{Z}}$ encoding the dynamical information that h can see. Whether it is a faithful image of X depends on how much the delay orbit $\{h \circ T^n : n \in \mathbb{Z}\}$ collectively distinguishes points of X . Injectivity of Φ_h modulo μ -null sets is equivalent to the observable σ -algebra

$$\mathcal{O}_h = \sigma(\{h \circ T^n : n \in \mathbb{Z}\})$$

equalling $\mathcal{B}(X)$ modulo μ .

The Koopman–Perron duality (Theorem 3.2) connects this to a function-space question: the observable σ -algebra $\mathcal{O}_h$ determines the Markov semigroup through the algebra it generates, and the reconstruction question is precisely whether that algebra is large enough to recover all of $\mathcal{B}$.

4.1 Separation defect and the role of CE

For a finite joined query algebra $\mathcal{G} = \mathcal{E}_{n_1} \vee \dots \vee \mathcal{E}_{n_k}$ from a directed system $(\mathcal{E}_n)_n$, the *separation defect* is

$$\delta(\mathcal{G}) := \sup_{A \in \sigma(\mathcal{C})} \inf_{E \in \mathcal{G}} \mu(A \triangle E),$$

where $\sigma(\mathcal{C})$ is the observable σ -algebra of the full system. It vanishes iff $\sigma(\mathcal{G}) = \sigma(\mathcal{C}) \bmod \mu$. A sequence $(\mathcal{G}_k)$ is *honest* if it is increasing and $\sigma(\bigcup_k \mathcal{G}_k) = \sigma(\mathcal{C}) \bmod \mu$; under an honest scheme $\delta(\mathcal{G}_k)$ is non-increasing.

Proposition 4.1 (CE drives separation defect to zero). *Let $(\mathcal{E}_n)_n$ be a query system with realization measure μ satisfying collective exhaustion [5]. Under any honest refinement scheme $(\mathcal{G}_k)$,*

$$\delta(\mathcal{G}_k) \longrightarrow 0 \quad \text{as } k \rightarrow \infty.$$

Proof. Fix $A \in \sigma(\mathcal{C})$. Since $(\mathcal{G}_k)$ is honest, $\sigma(\bigcup_k \mathcal{G}_k) = \sigma(\mathcal{C}) \bmod \mu$, so A is $\sigma(\bigcup_k \mathcal{G}_k)$ -measurable. By the $L^1(\mu)$ martingale convergence theorem, $\mathbf{1}_{E_k} \rightarrow \mathbf{1}_A$ in $L^1(\mu)$ where $E_k = \mathbb{E}_\mu[\mathbf{1}_A \mid \mathcal{G}_k]$ rounded to the nearest atom; in particular $\mu(A \triangle E_k) \rightarrow 0$. CE [5] ensures no mass persists on the vanishing sets $A \triangle E_k$, confirming $\inf_{E \in \mathcal{G}_k} \mu(A \triangle E) \rightarrow 0$ uniformly over A . Hence $\delta(\mathcal{G}_k) \rightarrow 0$. $\square$

Sections 5–6 carry out the reconstruction in the delay-map setting.

Remark 4.2 (The temporal specialisation). With $Q_t = h \circ T^t$, the algebra $\mathcal{O}_h^{(L)}$ is an honest refinement scheme once reconstruction holds (§§5–6). The separation defect specialises to $\delta(L) := \delta(\mathcal{O}_h^{(L)})$, the central quantity of [4]: exponential decay of $\delta(L)$ under mixing yields minimax-optimal certification rates. In the general query setting, exhaustion and mixing are separate conditions; the delay-map setting collapses the first by construction.

5 Density bridge and reconstruction

5.1 Density bridge lemma

Throughout this section and the next, $(X, \mathcal{B}, \mu)$ is a standard probability space, $T : X \rightarrow X$ is a measurable, μ -preserving invertible bimeasurable bijection with $T_*\mu = \mu$, and $h \in L^\infty(X, \mu)$.

Definition 5.1 (Observable algebra). The *observable algebra* generated by h is

$$\mathcal{O}_h := \sigma(\{h \circ T^n : n \in \mathbb{Z}\}) \subseteq \mathcal{B}(X).$$

The *delay algebra* $\mathcal{A}_h$ is the smallest subalgebra of $L^\infty(X, \mu)$ containing 1 and every $h \circ T^n$; concretely, finite sums $\sum_\alpha c_\alpha \prod_k (h \circ T^{n_k})^{e_k}$.

Definition 5.2 (Delay map). The *delay map* is

$$\Phi_h : X \rightarrow \mathbb{R}^{\mathbb{Z}}, \quad \Phi_h(x) = (h(T^n x))_{n \in \mathbb{Z}}.$$

Since $\pi_n \circ \Phi_h = h \circ T^n$ for each coordinate projection π_n , the map Φ_h is measurable and $\mathcal{O}_h = \Phi_h^{-1}(\mathcal{B}(\mathbb{R}^{\mathbb{Z}}))$. Thus $\mathcal{O}_h = \mathcal{B} \bmod \mu$ is precisely the condition that Φ_h is a measure-theoretic embedding.

The central lemma connects the Boolean algebra level to the L^2 level. It holds for any subalgebra of L^∞ .

Lemma 5.3 (Density bridge). *Let $(X, \mathcal{B}, \mu)$ be a probability space and let $\mathcal{A} \subseteq L^\infty(X, \mu)$ be a subalgebra containing the constant function 1. Then:*

$$\overline{\mathcal{A}}^{L^2} = L^2(X, \sigma(\mathcal{A}), \mu).$$

Consequently, $\overline{\mathcal{A}}^{L^2} = L^2(X, \mu)$ if and only if $\sigma(\mathcal{A}) = \mathcal{B}$ modulo μ .

Proof. Write $\mathcal{M} := \sigma(\mathcal{A})$ and $V := \overline{\mathcal{A}}^{L^2}$.

Step 1: $V \subseteq L^2(X, \mathcal{M}, \mu)$. Every $f \in \mathcal{A}$ is $\mathcal{A}$ -measurable, hence $\mathcal{M}$ -measurable. So $\mathcal{A} \subseteq L^2(X, \mathcal{M}, \mu)$. Since $L^2(X, \mathcal{M}, \mu)$ is a closed subspace of $L^2(X, \mathcal{B}, \mu)$, taking the closure gives $V \subseteq L^2(X, \mathcal{M}, \mu)$.

Step 2: $L^2(X, \mathcal{M}, \mu) \subseteq V$. It suffices to show that every indicator $\mathbf{1}_E$ with $E \in \mathcal{M}$ lies in V . By the functional monotone class theorem [1], the class

$$\mathcal{H} := \{f \in L^\infty(X, \mathcal{M}, \mu) : f \in V\}$$

is a vector space containing $\mathcal{A}$ and closed under bounded monotone pointwise limits (since if $f_n \in V$ are uniformly bounded and $f_n \rightarrow f$ pointwise a.e. with $f \in L^\infty$, then $f_n \rightarrow f$ in L^2 by dominated convergence, so $f \in V$). Since $\mathcal{A}$ is an algebra containing 1 and $\sigma(\mathcal{A}) = \mathcal{M}$, the monotone class theorem gives $\mathcal{H} = L^\infty(X, \mathcal{M}, \mu)$. In particular every $\mathcal{M}$ -indicator $\mathbf{1}_E \in V$.

Since the $\mathcal{M}$ -simple functions are dense in $L^2(X, \mathcal{M}, \mu)$ and V is closed, $L^2(X, \mathcal{M}, \mu) \subseteq V$.

Conclusion. Steps 1 and 2 give $V = L^2(X, \mathcal{M}, \mu)$. The final equivalence is immediate: $V = L^2(X, \mathcal{B}, \mu)$ iff $L^2(X, \mathcal{M}, \mu) = L^2(X, \mathcal{B}, \mu)$ iff $\mathcal{M} = \mathcal{B} \bmod \mu$. $\square$

Remark 5.4 (Role of the algebra structure). The algebra structure of $\mathcal{A}$ is essential. The corresponding statement for the closed *linear* span $\mathcal{H}_\mathcal{A} = \overline{\text{span}}(\mathcal{A})$ is false in general: $\mathcal{H}_\mathcal{A} = L^2$ does not imply $\sigma(\mathcal{A}) = \mathcal{B}$.

For example, take $X = [0, 1]$, $\mu = \text{Lebesgue}$, $T(x) = 2x \bmod 1$ (the doubling map), and $h(x) = x$. The orbit $\{h \circ T^n\}(x) = \{2^n x \bmod 1\}$ generates $\mathcal{B}([0, 1])$ as a σ -algebra (the dyadic structure separates points). But $\text{span}\{2^n x \bmod 1 : n \geq 0\}$ is a proper subspace of $L^2[0, 1]$: it does not contain x^2 . So the linear span can be a proper dense subspace even when $\sigma(\mathcal{A}) = \mathcal{B}$.

In the other direction, h cyclic for U_T (linear span dense) always implies $\sigma(\mathcal{A}_h) = \mathcal{B} \bmod \mu$ — see Corollary 6.2.

5.2 Reconstruction theorem

Theorem 5.5 (Reconstruction theorem). *Let $(X, \mathcal{B}, \mu, T)$ be an invertible measure-preserving system and $h \in L^\infty(X, \mu)$. The following are equivalent:*

- (i) $\mathcal{O}_h = \mathcal{B}$ modulo μ ;
- (ii) $\mathcal{A}_h$ is dense in $L^2(X, \mu)$;
- (iii) $\Phi_h : X \rightarrow \mathbb{R}^\mathbb{Z}$ is a measure-theoretic embedding: $\Phi_h^{-1}(\mathcal{B}(\mathbb{R}^\mathbb{Z})) = \mathcal{B}$ modulo μ .

Proof. (i) $\Leftrightarrow$ (ii). Apply the density bridge (Lemma 5.3) with $\mathcal{A} = \mathcal{A}_h$ and $\sigma(\mathcal{A}_h) = \mathcal{O}_h$: $\overline{\mathcal{A}_h}^{L^2} = L^2(X, \mu)$ iff $\mathcal{O}_h = \mathcal{B} \bmod \mu$.

(i) $\Leftrightarrow$ (iii). By Definition 5.2, $\mathcal{O}_h = \Phi_h^{-1}(\mathcal{B}(\mathbb{R}^\mathbb{Z}))$. So $\mathcal{O}_h = \mathcal{B} \bmod \mu$ iff $\Phi_h^{-1}(\mathcal{B}(\mathbb{R}^\mathbb{Z})) = \mathcal{B} \bmod \mu$, which is exactly condition (iii). $\square$

Remark 5.6 (What “measure-theoretic embedding” means). Condition (iii) says that Φ_h is *almost everywhere injective* and the pullback σ -algebra is all of $\mathcal{B}$. More precisely: since $\Phi_h^{-1}(\mathcal{B}(\mathbb{R}^\mathbb{Z})) = \mathcal{B} \bmod \mu$, for any two sets $A, A' \in \mathcal{B}$ with $\mu(A \triangle A') > 0$ there exists a Borel set $E \subseteq \mathbb{R}^\mathbb{Z}$ with $\mu(\Phi_h^{-1}(E) \triangle A) = 0$ and $\mu(\Phi_h^{-1}(E) \triangle A') > 0$. In measure-theoretic language, Φ_h induces an isomorphism of measure algebras $(X/\mu, \mathcal{B}/\mu) \cong (\mathbb{R}^\mathbb{Z}/(\Phi_h)_*\mu, \mathcal{B}(\mathbb{R}^\mathbb{Z})/(\Phi_h)_*\mu)$.

The T -invariance of μ implies that $(\Phi_h)_*\mu$ is invariant under the bilateral shift σ on $\mathbb{R}^\mathbb{Z}$, since $(\Phi_h(Tx))_n = h(T^{n+1}x) = (\sigma(\Phi_h(x)))_n$ for each n . The dynamical system (X, T, μ) is therefore isomorphic to $(\mathbb{R}^\mathbb{Z}, \sigma, (\Phi_h)_*\mu)$ restricted to the image of Φ_h .

6 Cyclic vectors and Stone space identification

6.1 Cyclic vectors

Definition 6.1 (Cyclic vector). The function $h \in L^2(X, \mu)$ is a *cyclic vector* for U_T if

$$\mathcal{H}_h := \overline{\text{span}}\{U_T^n h : n \in \mathbb{Z}\} = L^2(X, \mu).$$

Corollary 6.2 (Cyclic implies reconstruction). *If $h \in L^\infty(X, \mu)$ is a cyclic vector for U_T , then $\mathcal{O}_h = \mathcal{B}$ modulo μ .*

Proof. Since $\mathcal{A}_h$ contains $\text{span}\{h \circ T^n\} = \text{span}\{U_T^n h\}$ as a subset, $\overline{\mathcal{A}_h}^{L^2} \supseteq \mathcal{H}_h = L^2$. By the density bridge, $\mathcal{O}_h = \mathcal{B} \bmod \mu$. $\square$

Remark 6.3 (The implication is strict). The converse of Corollary 6.2 fails in general: $\mathcal{O}_h = \mathcal{B}$ does not imply h is cyclic. The doubling map example of Remark 5.4 witnesses this.

The cyclic vector condition asks for density of the *linear span* of the orbit; reconstruction asks only for density of the *algebra*. For a bounded h , the linear span is contained in the algebra, so cyclicity is the stronger condition.

The two conditions coincide precisely when U_T has *simple L^2 -spectrum*: $L^2(X, \mu)$ is a cyclic U_T -module, meaning there exists some cyclic vector. In this case, every generating observation h (satisfying $\mathcal{O}_h = \mathcal{B}$) is automatically cyclic. Simple spectrum is a generic property of ergodic transformations in the Baire category sense [2] but is not universal.

6.2 Stone space identification

The Stone space $\text{St}(\mathcal{O}_h)$ of the observable algebra is the compact Hausdorff space whose points are ultrafilters on $\mathcal{O}_h$, with the natural Stone embedding $\iota : X \rightarrow \text{St}(\mathcal{O}_h)$ sending x to the principal ultrafilter $\{E \in \mathcal{O}_h : x \in E\}$.

Proposition 6.4 (Stone space identification). *Suppose $\mathcal{O}_h = \mathcal{B}$ modulo μ . Then ι is a measure-theoretic isomorphism: ι identifies the measure algebra $(X/\mu, \mathcal{B}/\mu)$ with $(\text{St}(\mathcal{O}_h)/\iota_*\mu, \mathcal{B}(\text{St}(\mathcal{O}_h))/\iota_*\mu)$.*

Proof. Injectivity mod μ . Suppose $\iota(x) = \iota(x')$. For any $E \in \mathcal{B}$ with $x \in E \not\equiv x'$, find $F \in \mathcal{O}_h$ with $\mu(E \triangle F) = 0$. Since $x \in F$ and $\iota(x) = \iota(x')$, also $x' \in F$, so $x' \in (E \triangle F) \cup F^c$, placing x' in a null set. Hence ι is injective μ -a.e.

σ -algebra identification. The pullback ι^{-1} maps the Baire σ -algebra of $\text{St}(\mathcal{O}_h)$ (generated by the clopen sets $[E] = \{u : E \in u\}$) to $\mathcal{O}_h = \mathcal{B} \bmod \mu$.

Measure recovery. Since μ is σ -additive on $\mathcal{O}_h = \mathcal{B}$, the collective exhaustion condition of [5] forces $\iota_*\mu$ to concentrate on the principal ultrafilters $\iota(X)$, so $\iota_*\mu$ is supported on the image and the measure algebras are identified. $\square$

The Stone space [5] — built as a compact ambient space for deriving σ -additivity — is, under the reconstruction condition, the state space X itself. The construction that began by seeking a compact completion of the sample space ends by recognising it as the object the observer was trying to recover.

Example 6.5 (Cantor set: reconstruction without smoothness). Let $\mathcal{C} \subset [0, 1]$ be the middle-thirds Cantor set, and let $\mu_{\mathcal{C}}$ be the Cantor measure — the pushforward of the fair-coin product measure $\nu = (\frac{1}{2}\delta_0 + \frac{1}{2}\delta_2)^{\mathbb{N}}$ under the ternary coding map $\pi(d_1, d_2, \dots) = \sum_{k \geq 1} d_k/3^k$. Let $S : \mathcal{C} \rightarrow \mathcal{C}$ be the shift map

$$S(x) = \begin{cases} 3x & x \in \mathcal{C} \cap [0, \frac{1}{3}], \\ 3x - 2 & x \in \mathcal{C} \cap [\frac{2}{3}, 1], \end{cases}$$

and let $h(x) = x$. The system $(\mathcal{C}, \mu_{\mathcal{C}}, S)$ is measurably conjugate to the Bernoulli shift $(\{0, 2\}^{\mathbb{N}}, \nu, \sigma)$ via π ; in particular S is $\mu_{\mathcal{C}}$ -invariant and ergodic.

For $x = \pi(d_1, d_2, \dots)$ we have $S^n x = \pi(d_{n+1}, d_{n+2}, \dots)$. Since every point of $\mathcal{C}$ lies in $[0, \frac{1}{3}] \cup [\frac{2}{3}, 1]$,

$$d_{n+1} = 0 \iff S^n x \in [0, \frac{1}{3}], \quad d_{n+1} = 2 \iff S^n x \in [\frac{2}{3}, 1],$$

so $h \circ S^n$ recovers the $(n+1)$ -st ternary digit of x . The delay vector $\Phi_h^{(L)}(x) = (x, Sx, \dots, S^L x)$ reads off digits $d_1, \dots, d_{L+1}$; its fibres are the generation- $(L+1)$ cylinder sets

$$F_z = \{x \in \mathcal{C} : d_k(x) = z_k, 1 \leq k \leq L+1\}, \quad z \in \{0, 2\}^{L+1},$$

each carrying $\mu_{\mathcal{C}}(F_z) = 2^{-(L+1)}$, so $\mathcal{O}_h^{(L)} = \sigma(\Phi_h^{(L)}) = \sigma\{F_z : z \in \{0, 2\}^{L+1}\}$. Since $\mu_{\mathcal{C}}$ -a.e. pair of distinct points differs at some finite digit, these algebras separate points as $L \rightarrow \infty$:

$$\mathcal{O}_h = \sigma\left(\bigcup_{L \geq 0} \mathcal{O}_h^{(L)}\right) = \mathcal{B}(\mathcal{C}) \bmod \mu_{\mathcal{C}}.$$

By Theorem 5.5, Φ_h is a measure-theoretic embedding: the finite-lag observable algebras refine to the full Borel structure as $L \rightarrow \infty$.

$\mathcal{C}$ is not a smooth manifold: it is compact and zero-dimensional, with Hausdorff dimension $\log 2 / \log 3 \approx 0.63$. The differential-geometric hypotheses of Takens's theorem — C^2 ambient dynamics, integer manifold dimension d , embedding bound $2d+1$ — are unavailable before one even asks about regularity of S . Takens has no domain of application here. Theorem 5.5 applies because it requires only that the observable algebra be measurably generated; smoothness plays no role.

The classical Takens embedding theorem [7] reaches a related conclusion by a different route: for a C^2 diffeomorphism on a compact d -manifold and a generic smooth h , the finite-delay map $\Phi_h^{(k)}(x) = (h(x), h(Tx), \dots, h(T^{k-1}x))$ is a diffeomorphic embedding into $\mathbb{R}^{2d+1}$ for $k \geq 2d + 1$. The reconstruction theorem here requires only measurability, uses all delays simultaneously, and replaces the dimension count with the algebraic density condition; in exchange, the embedding is measure-theoretic rather than topological. On a compact manifold with generic smooth h , the orbit $\{h \circ T^n\}$ separates points, so Stone–Weierstrass gives $\mathcal{A}_h$ uniformly dense in $C(M)$ and hence L^2 -dense: the density condition holds and the two results are compatible.

Reconstruction is an exact asymptotic condition: it asks whether the delay algebra generates all of $\mathcal{B}$ in the limit of all lags, and says nothing about rates, finite approximations, or what a finite time series can certify. Those questions are the subject of what follows.

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